

CNNs and UNets

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Building ML/AI applications
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Four main tasks in computer vision

Regression and
classification of images



Cat? Yes

Dog? Yes

Bat? No

Feature extraction



Cat color: black

Dog color: brown

Cat's eye color:
green

Dog's eye color:
brown

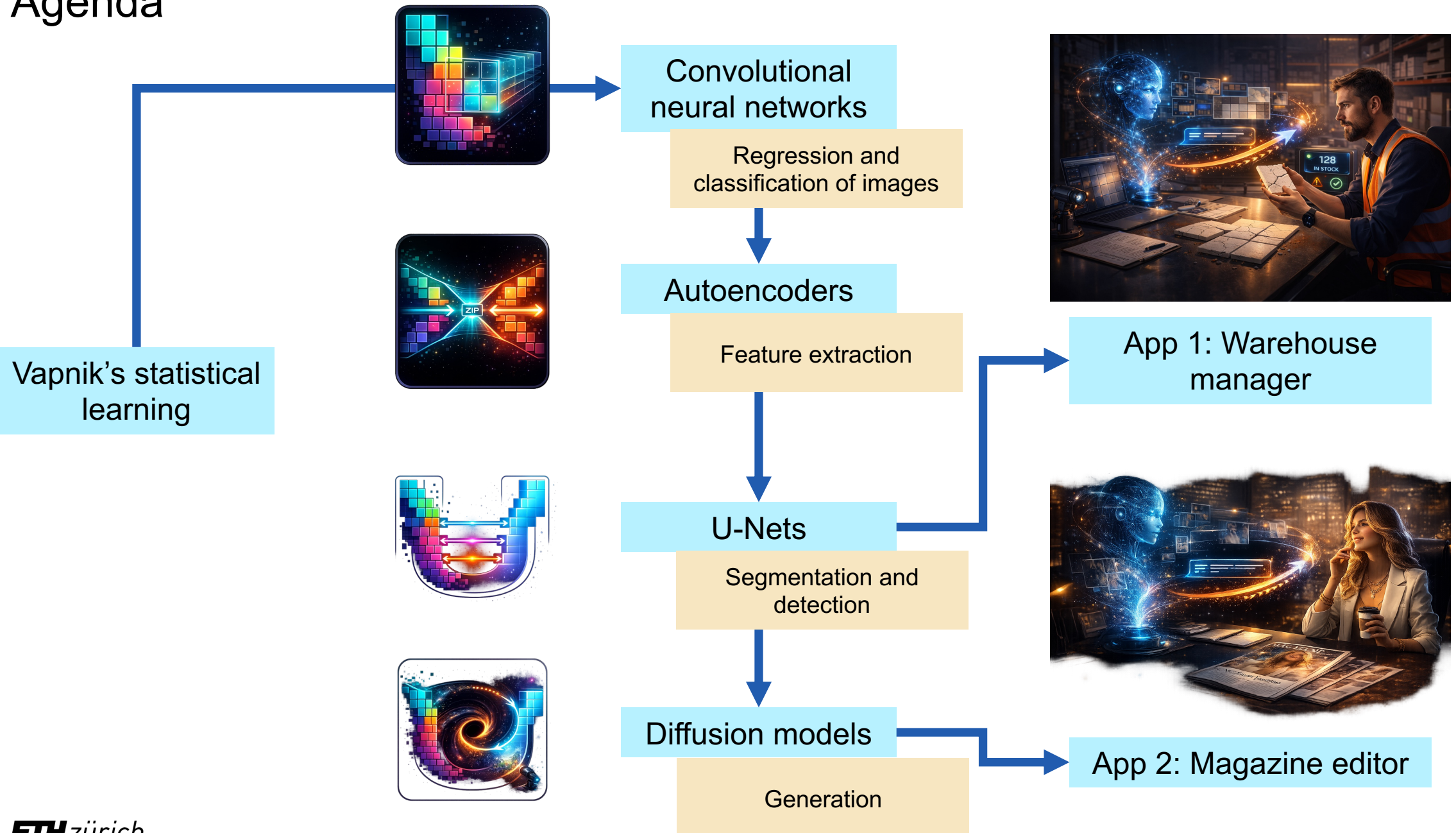
Segmentation and
detection



Generation

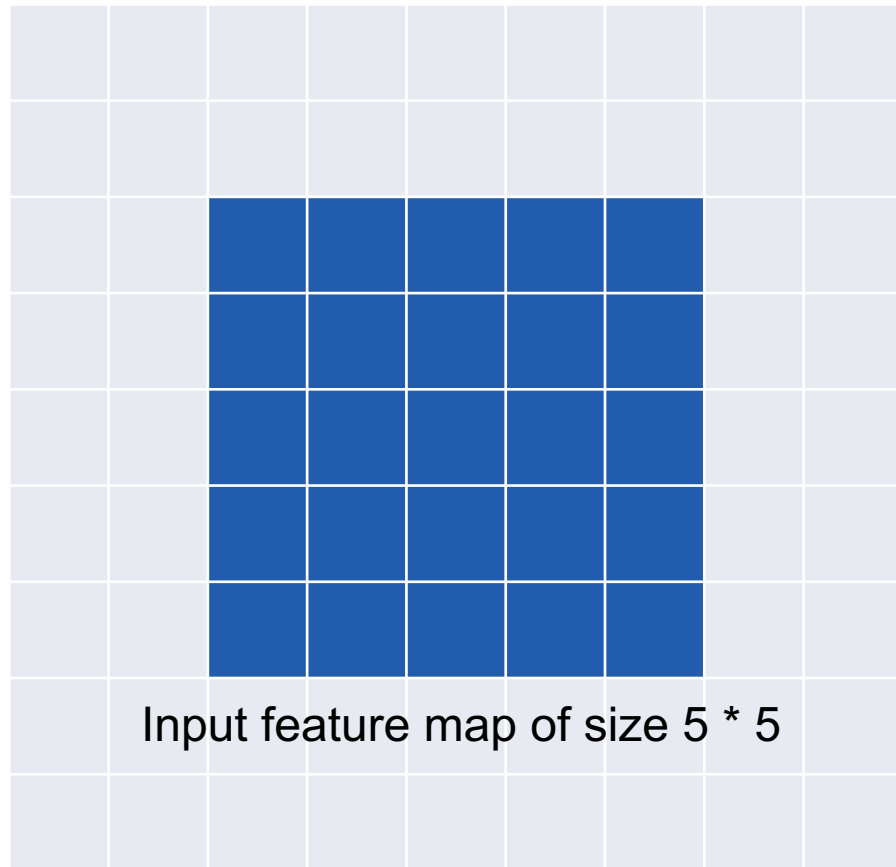


Agenda

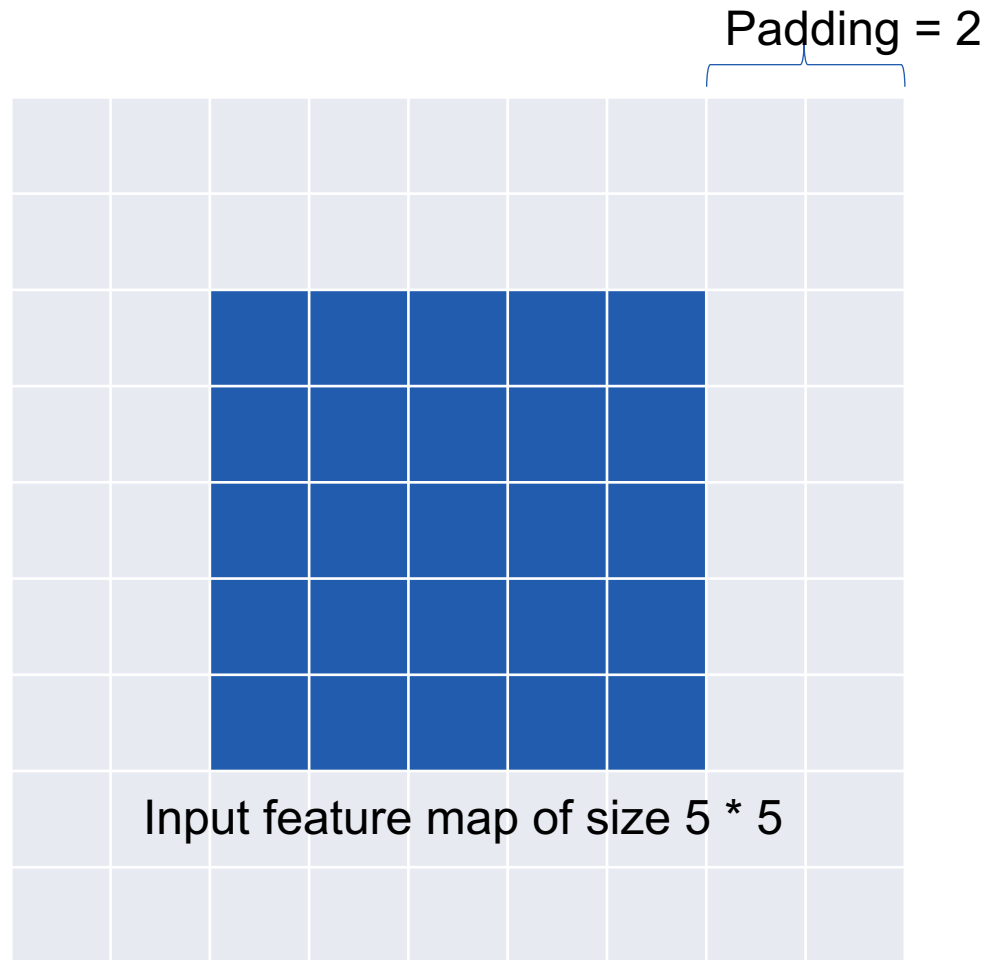


Convolutions and deconvolutions

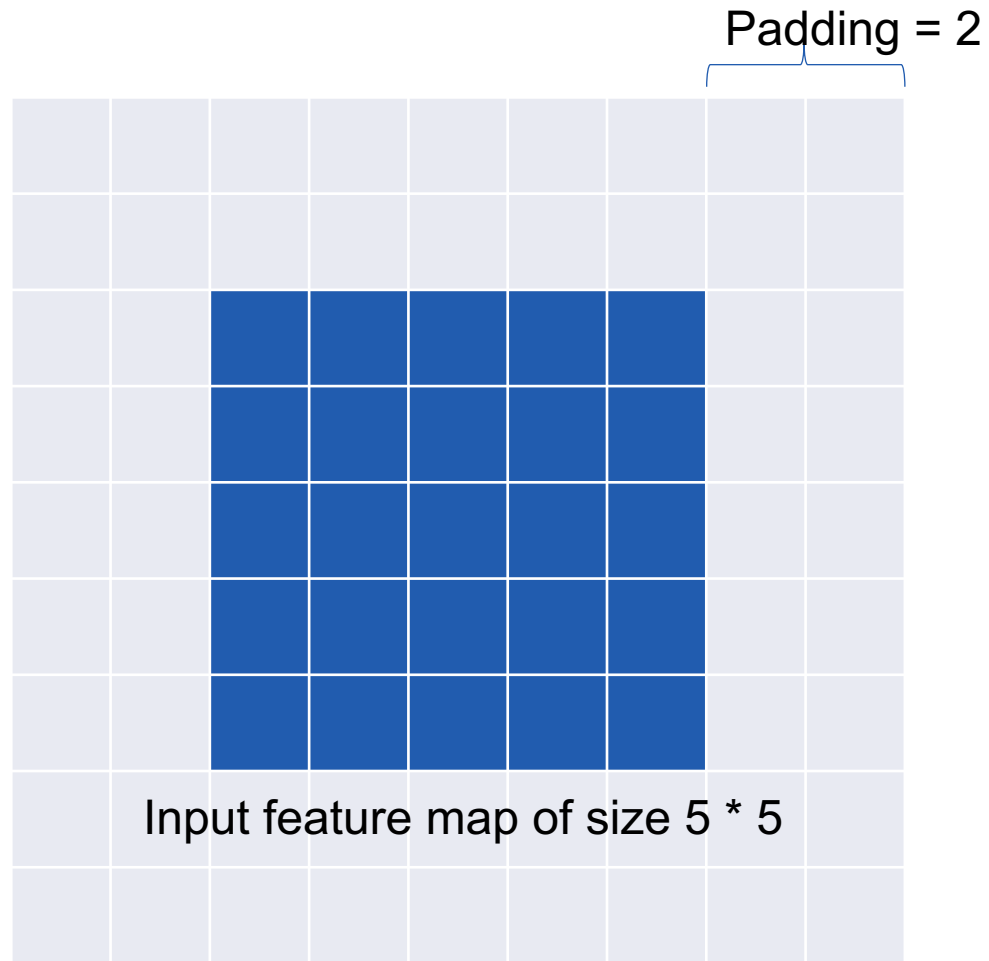
Convolutional filters



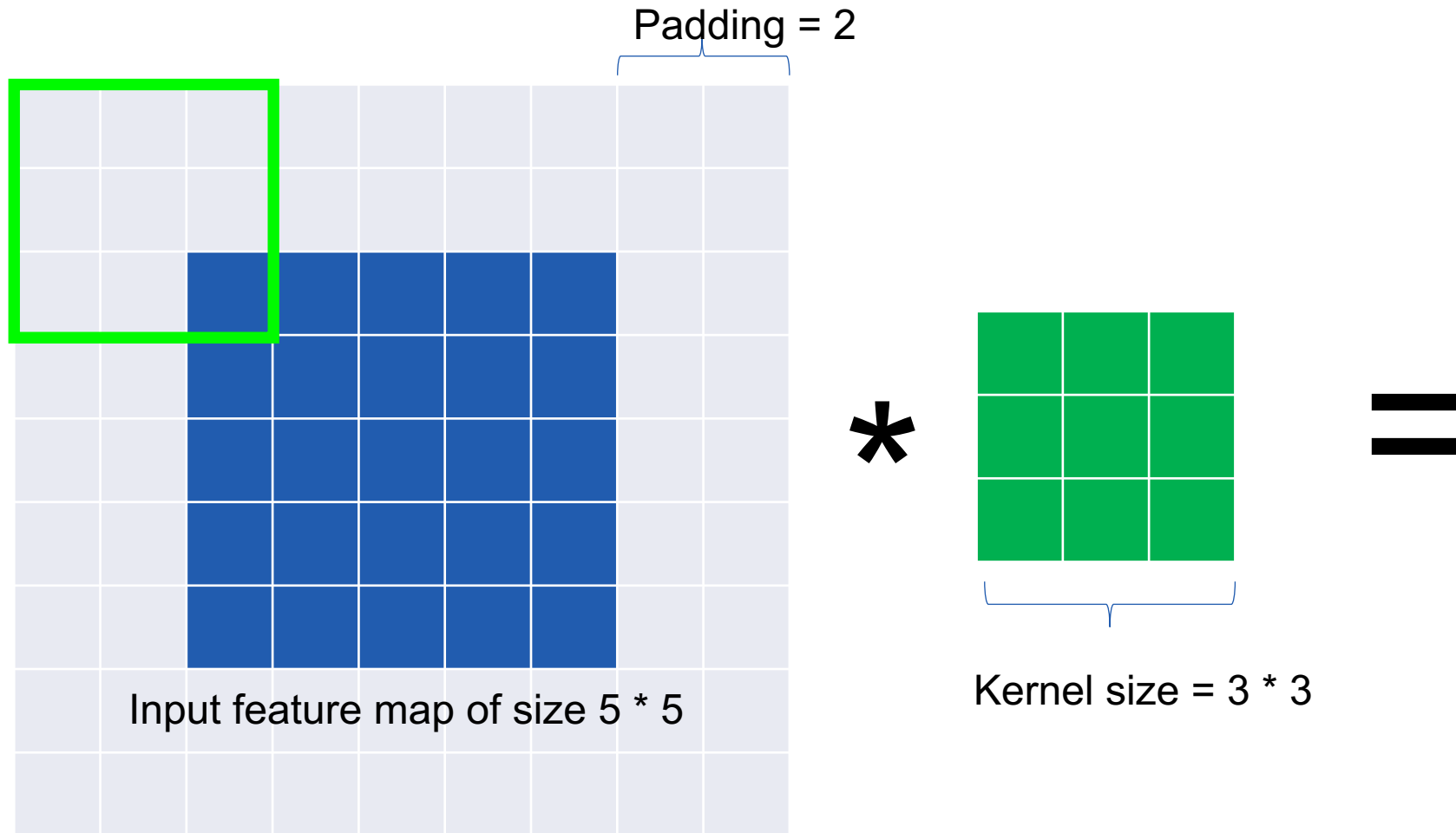
Convolutional filters



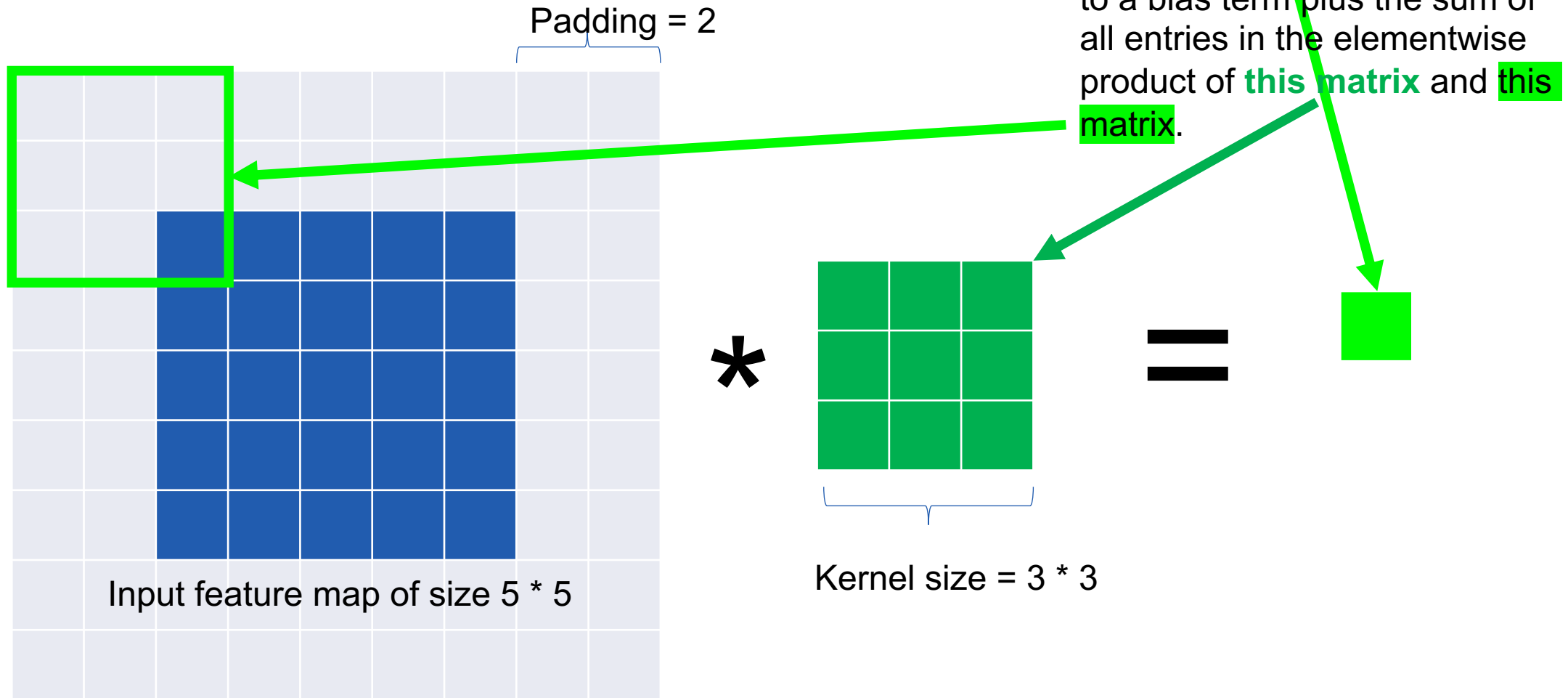
Convolutional filters



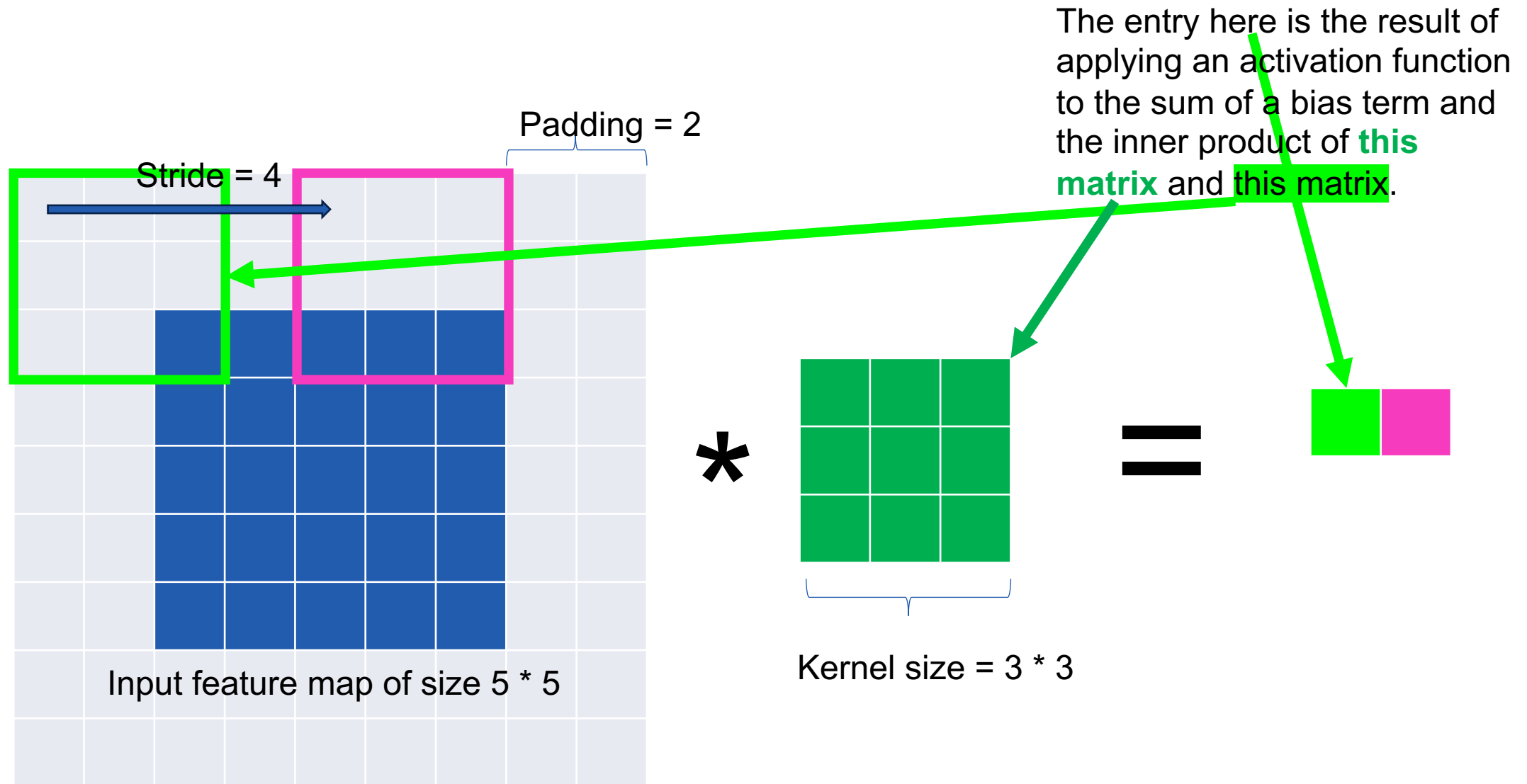
Convolutional filters



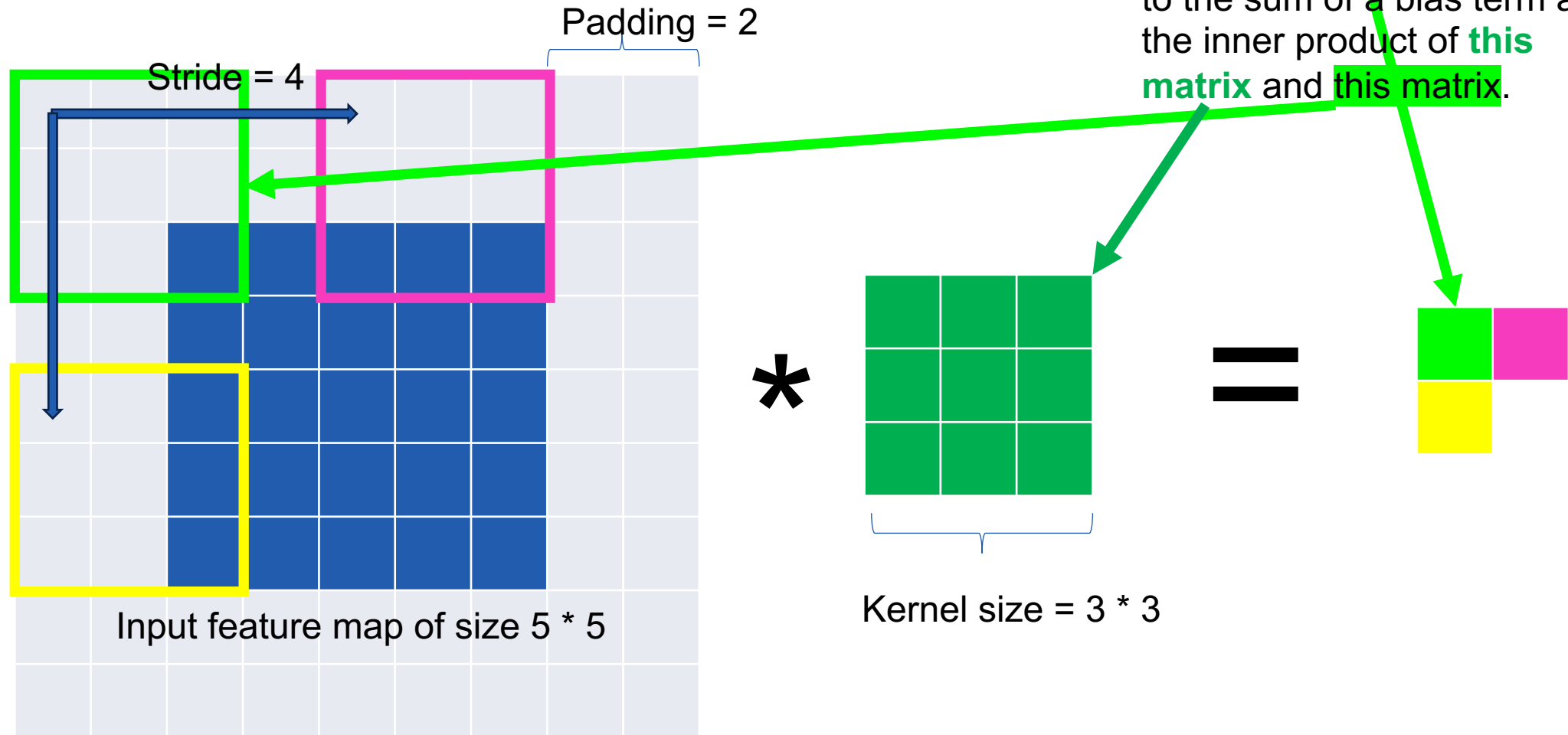
Convolutional filters



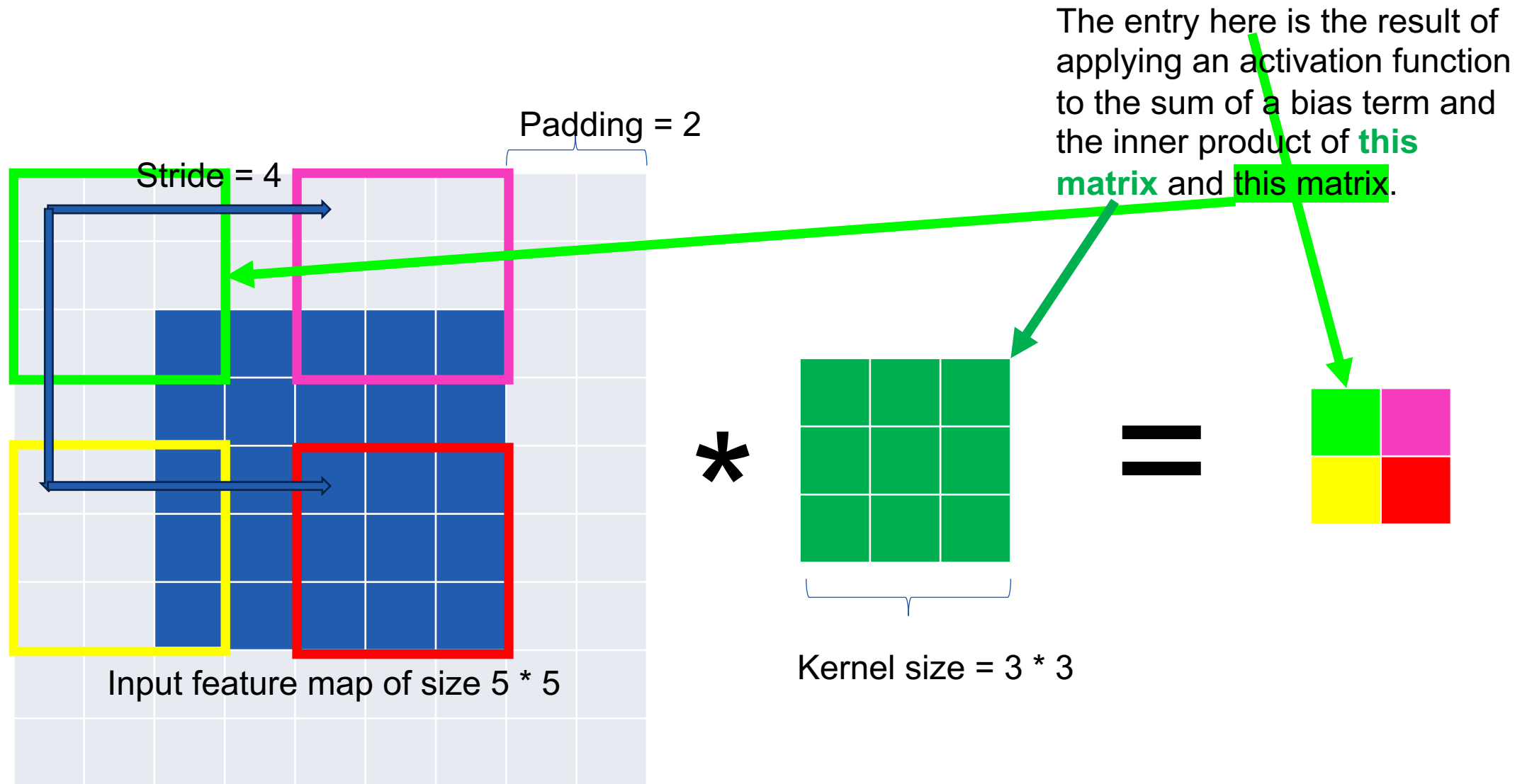
Convolutional filters



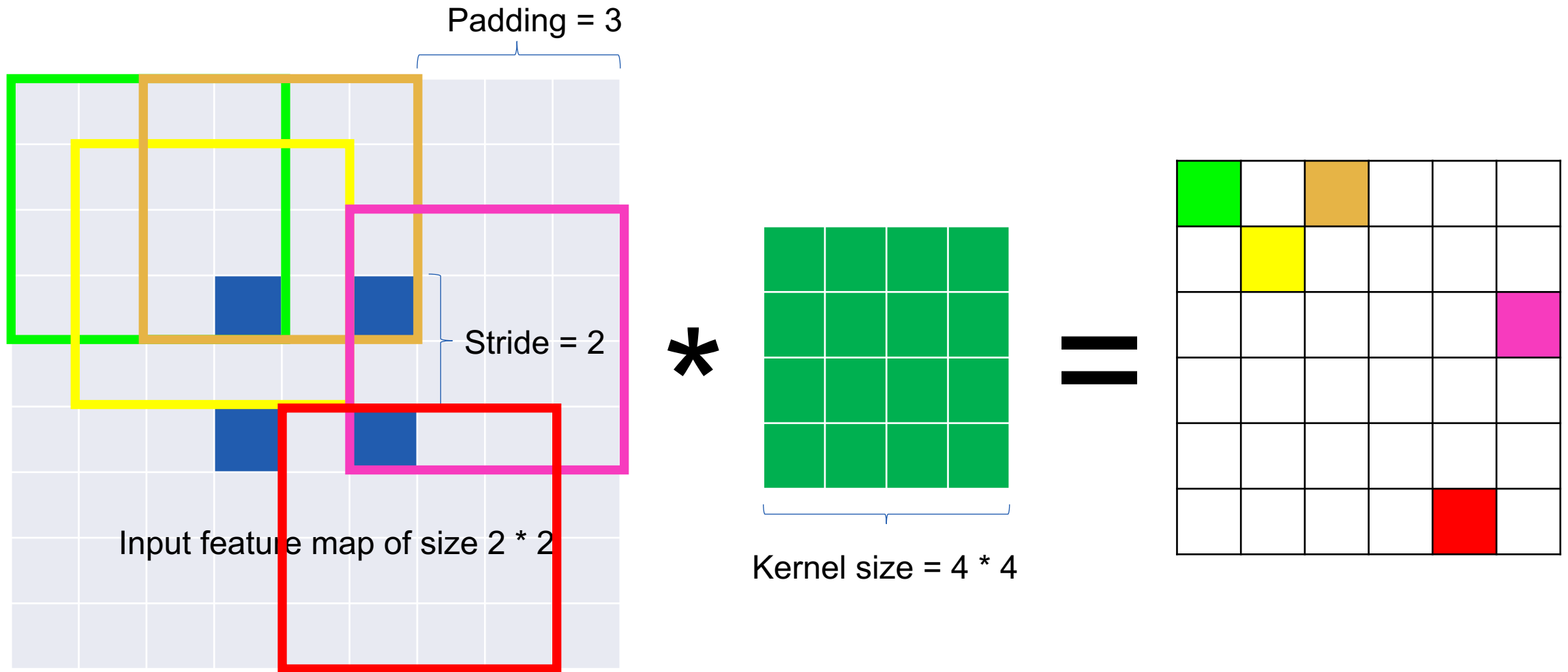
Convolutional filters



Convolutional filters

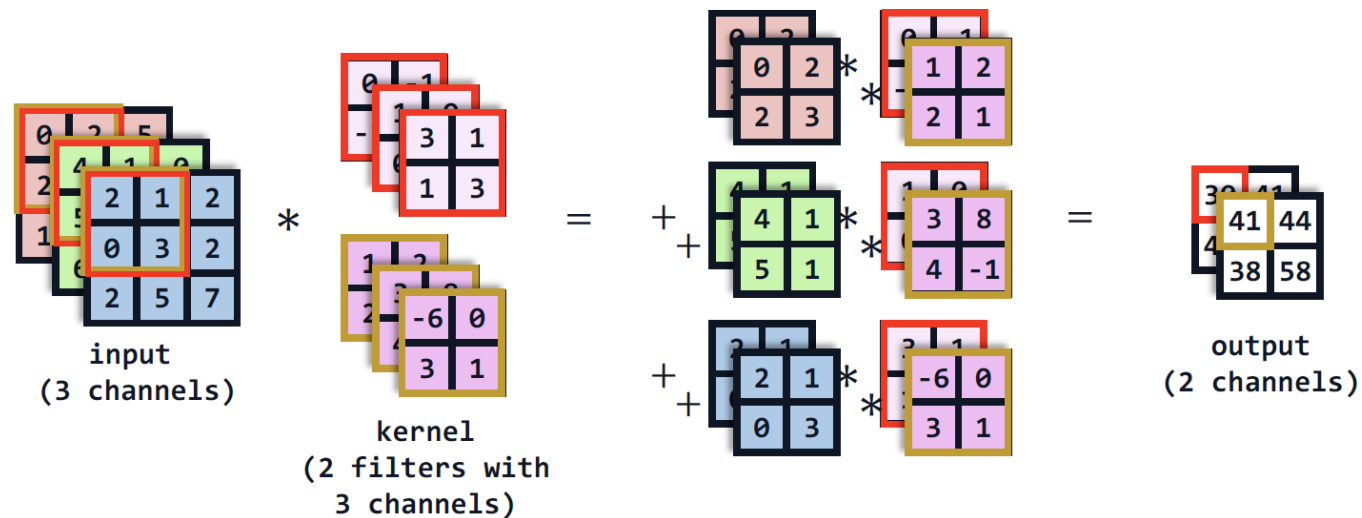


Deconvolutional filters



Convolutional and deconvolutional layers

- A (de)convolutional layer consists of a number of equally-sized (de)convolutional filters, a padding, and a stride.
- The output is a tensor where the number of channels matches the number of filters in the layer.



U-net

- A typical architecture for image segmentation.
- One can also treat a U-net as some sort of image autoencoder
- Image from original paper:
<https://arxiv.org/abs/1505.04597>

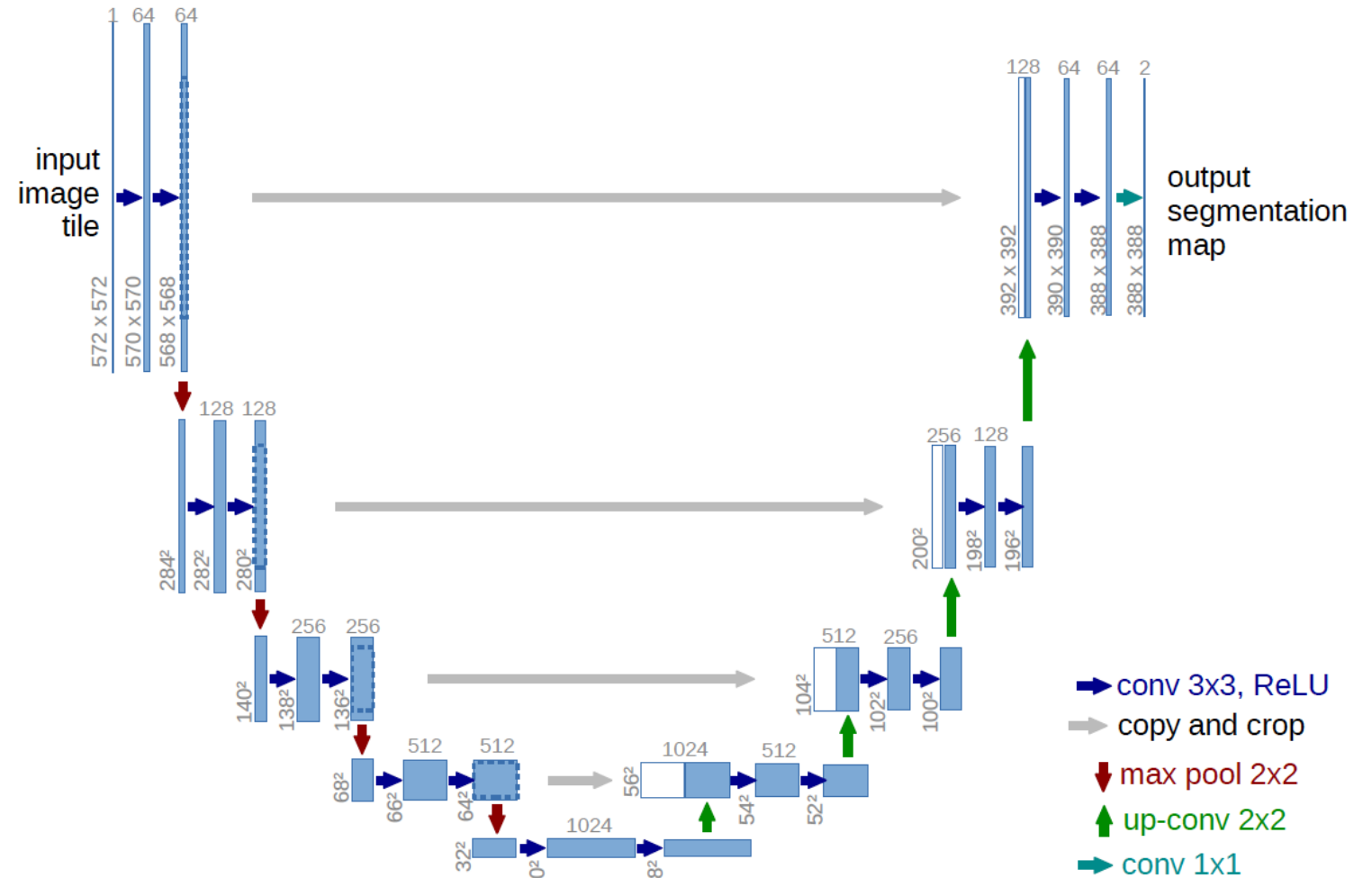
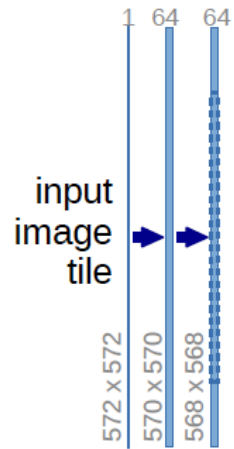


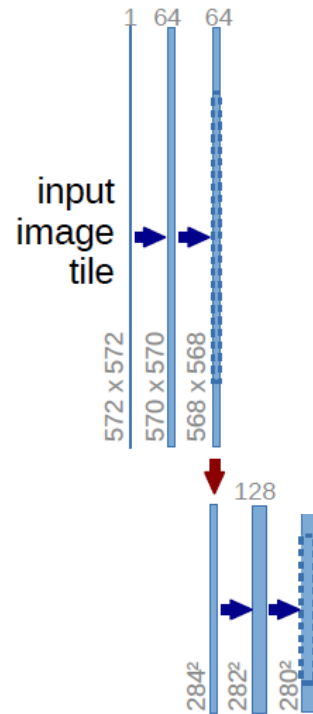
Fig. 1. U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

U-net



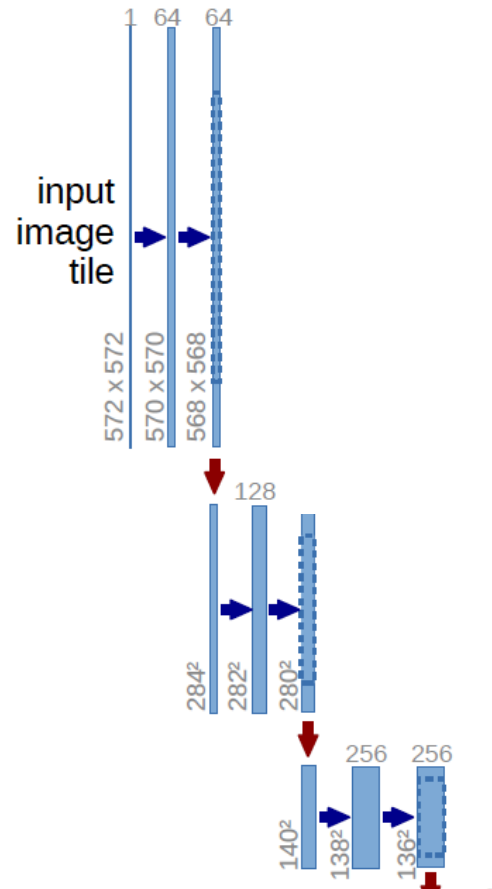
- conv 3x3, ReLU
- copy and crop
- ↓ max pool 2x2
- ↑ up-conv 2x2
- conv 1x1

U-net



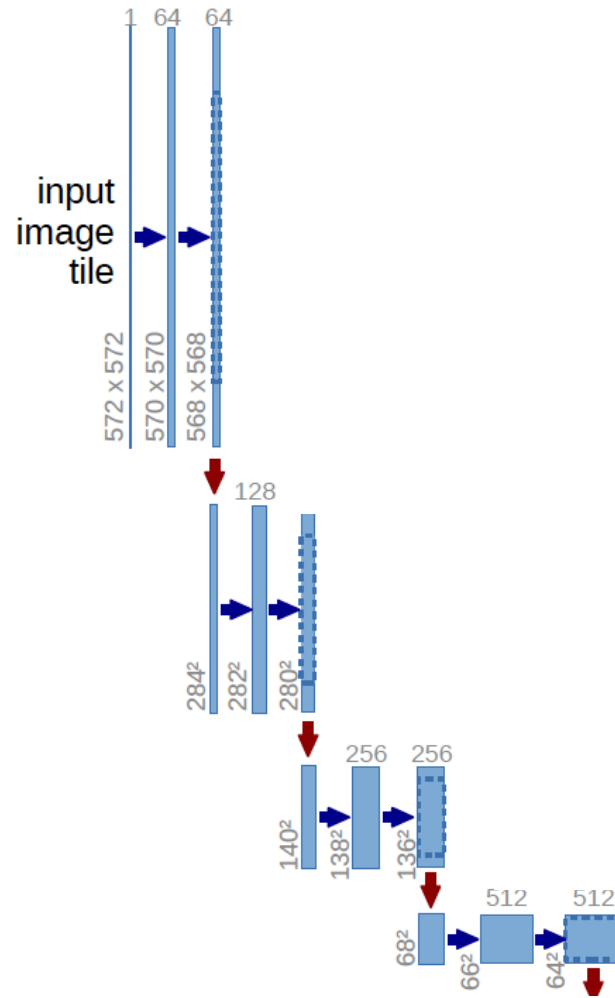
- conv 3x3, ReLU
- copy and crop
- ↓ max pool 2x2
- ↑ up-conv 2x2
- conv 1x1

U-net



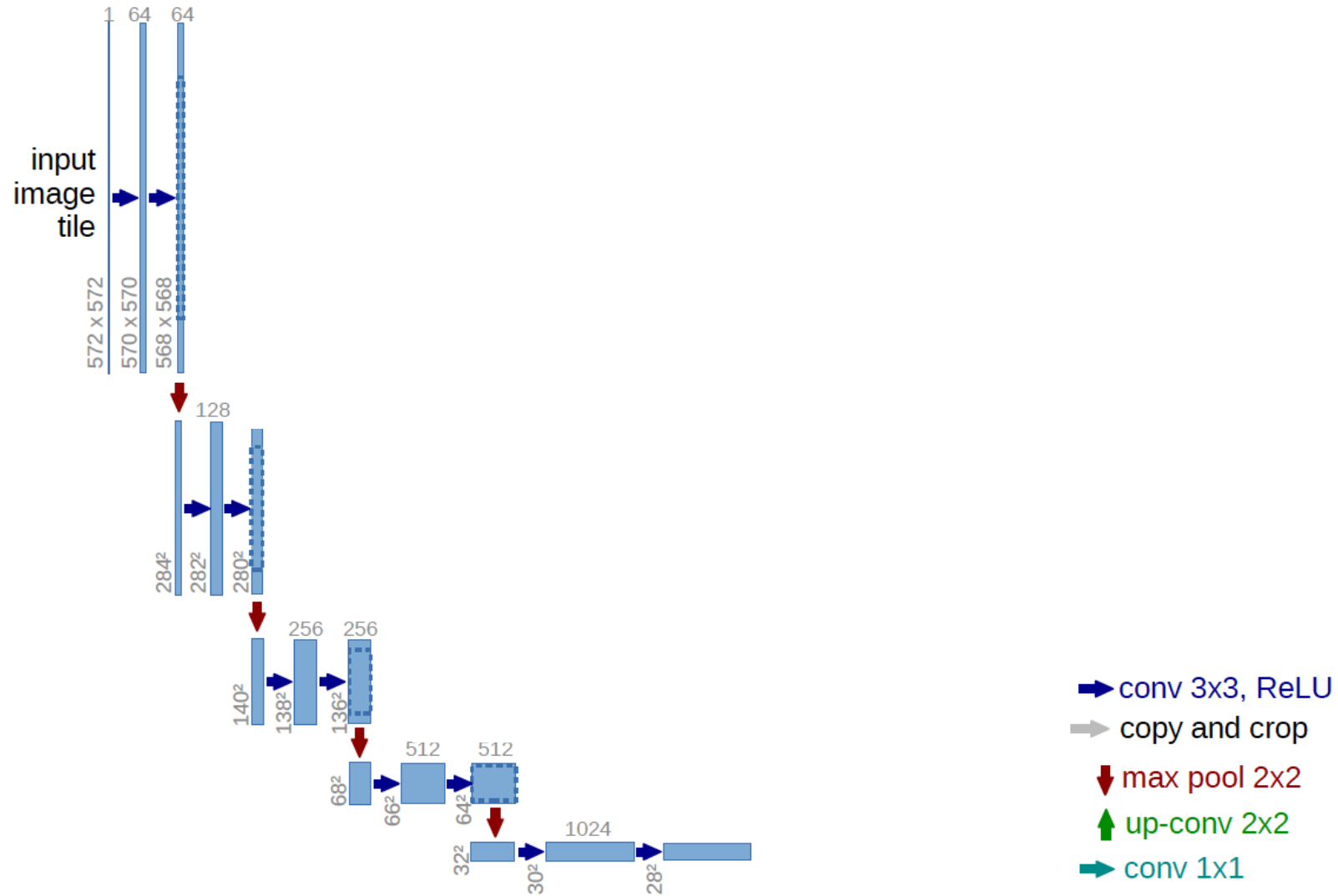
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U-net

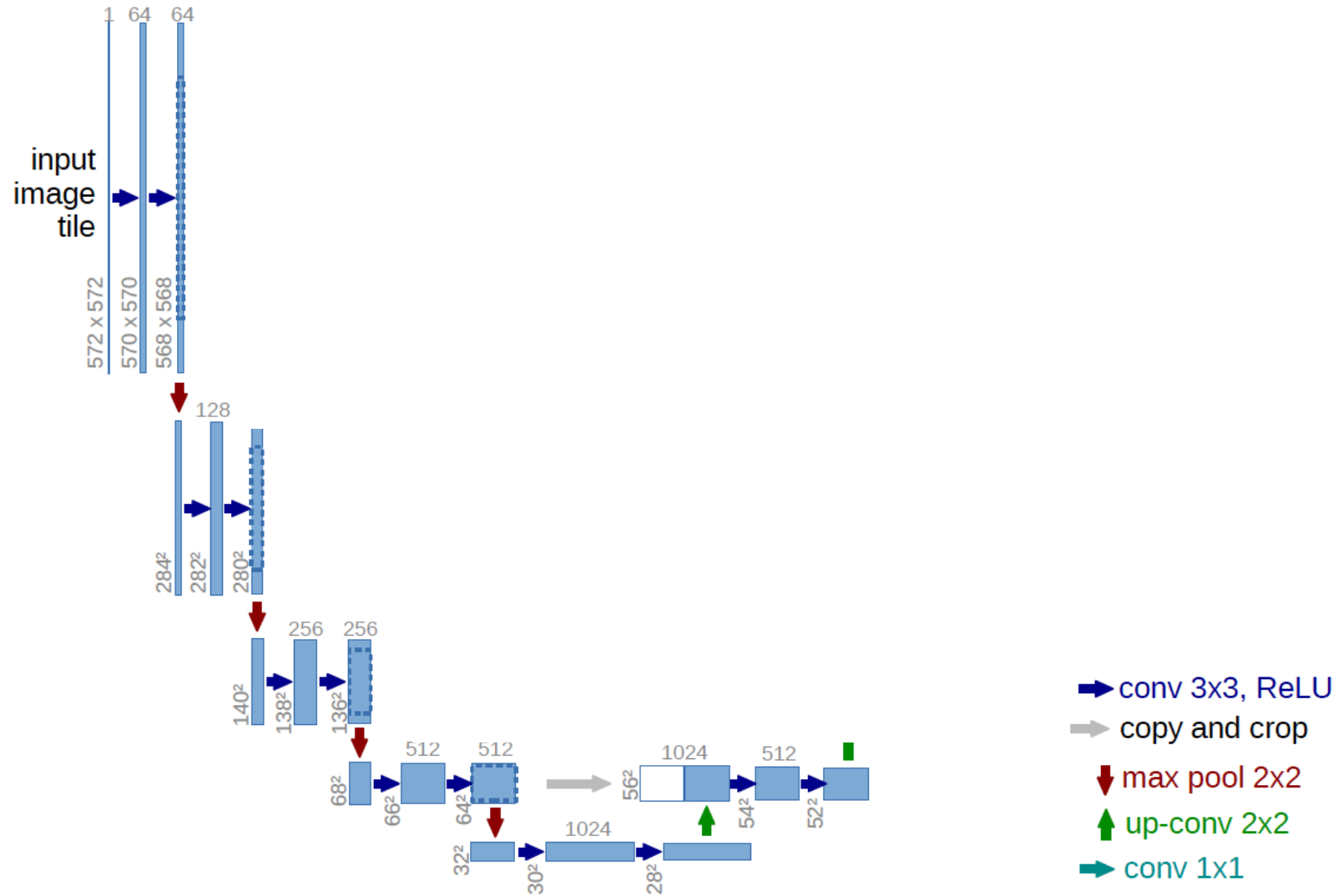


- conv 3x3, ReLU
- copy and crop
- ↓ max pool 2x2
- ↑ up-conv 2x2
- conv 1x1

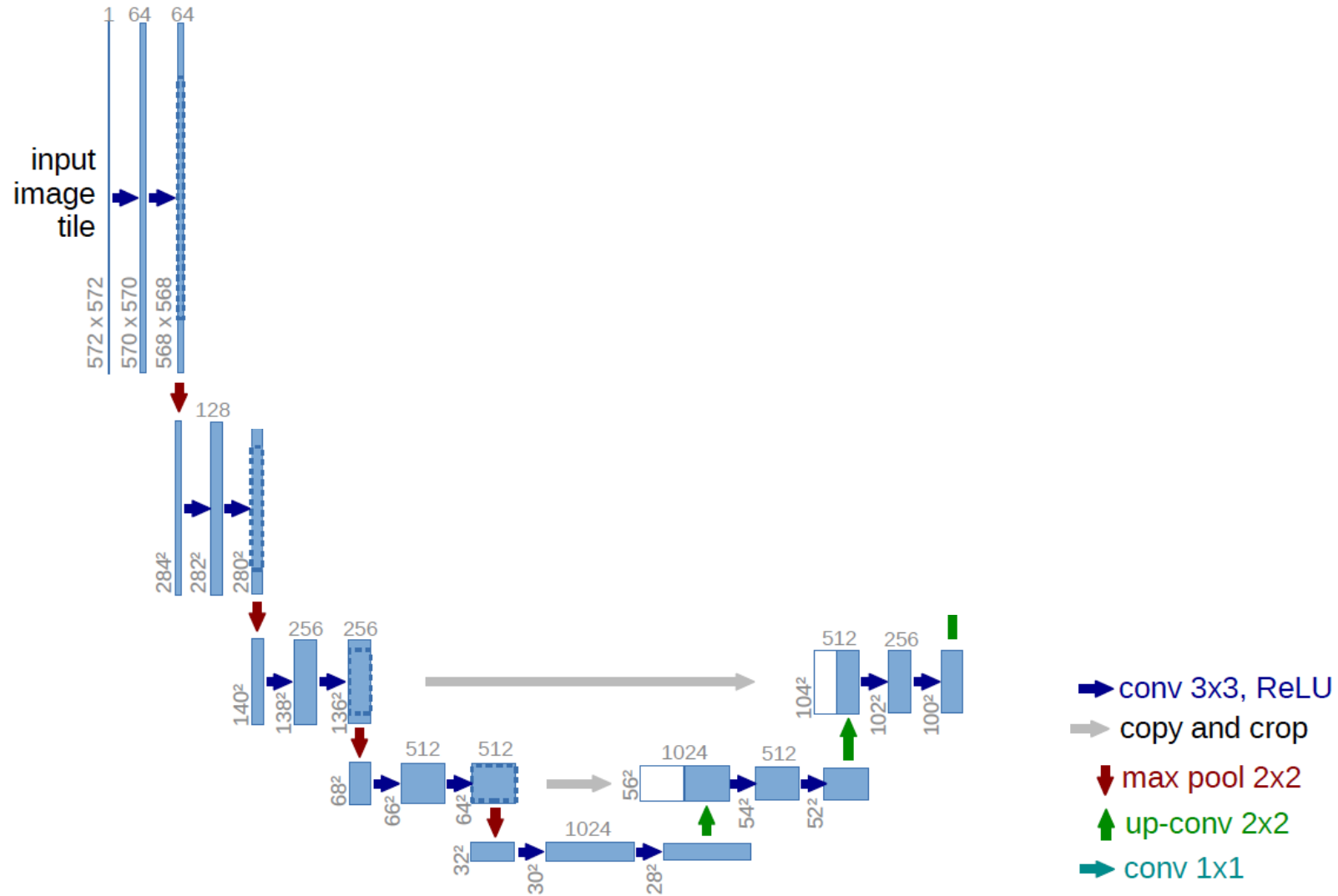
U-net



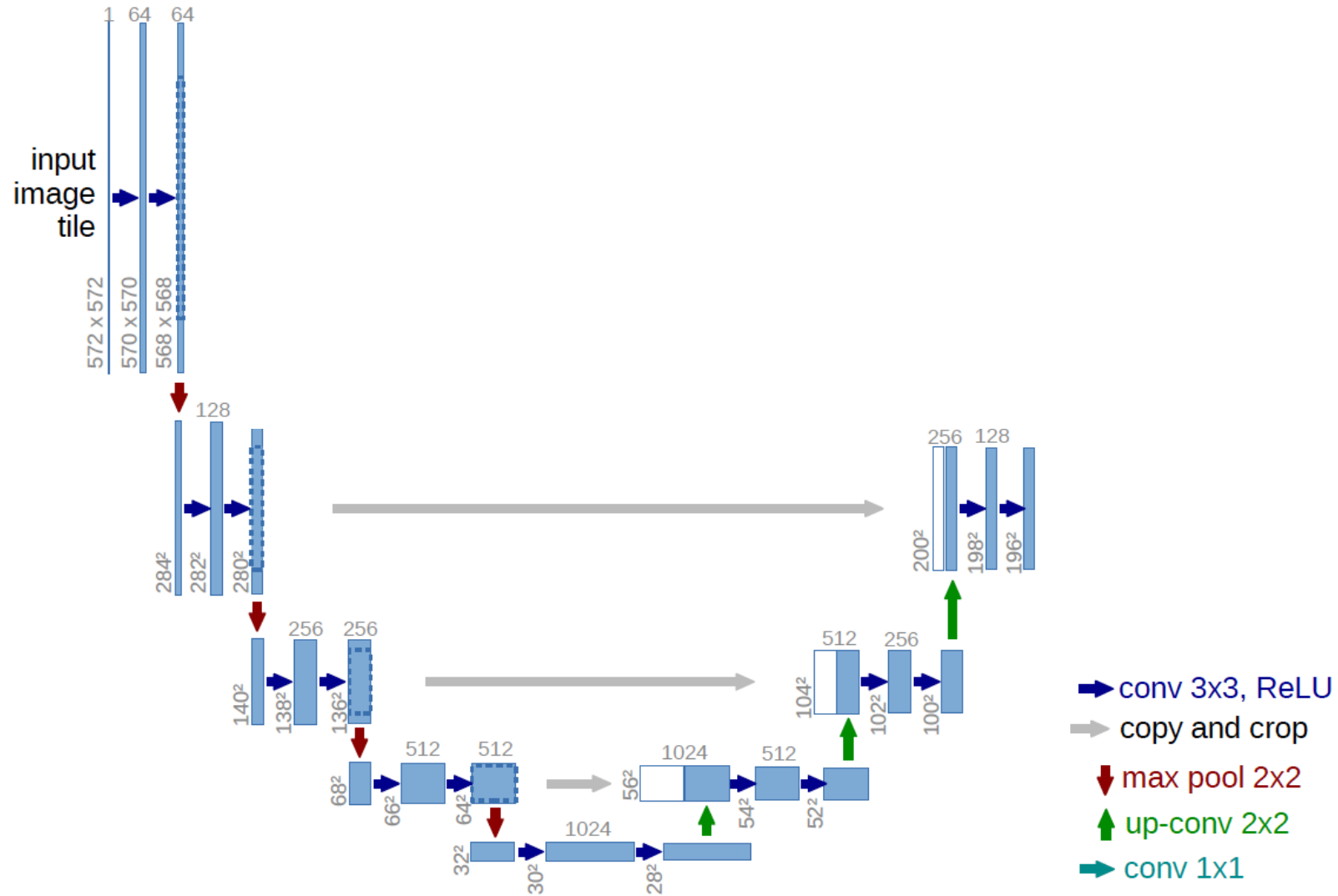
U-net



U-net



U-net



U-net

